CockpitSentinel: External Benchmarks Report

Performance on YawDD, Kaggle NTHU, and Kaggle State Farm Datasets

Benchmark Summary: Evaluated across 3 independent standard datasets. Achieved 100% video-level accuracy on YawDD, 86% accuracy on Kaggle NTHU nodding (99.2% YOLO specificity), and 99.7% phone recall on Kaggle State Farm.

1. YawDD Benchmark (Driver Video Fatigue Sessions)

Evaluated on 10 full driver sessions (1,162 evaluated frames, male/female, glasses, sunglasses):

Metric	Score	Description
Video-Level Accuracy	100.00%	All 10 driver sessions correctly identified
Yawn Detection Recall	100.00%	3 / 3 fatigue sessions caught without failure
Specificity (Talking/Normal)	100.00%	7 / 7 negative control sessions had 0 false alarms
F1-Score	100.00%	Clean separation between talking and yawning
Mean MAR in Normal Driving	0.015	Mouth closed, normal breathing
Mean MAR while Talking	0.148	Mouth moving, safely below 0.60 threshold
Peak MAR during Yawning	1.090	Prominent spike triggering alarm perimeter
Distraction Specificity	93.80%	0.0% phone/smoke false alarms during natural driving

2. Kaggle NTHU Driver Drowsiness Benchmark (250 Images)

Category	Tested State	Images	Accuracy	Avg EAR	Avg MAR
slowBlinkWithNodding	Head Pitch & Slow Blink	50	86.00%	0.151	0.082
sleepyCombination	Micro-sleep	50	38.00%	0.340	0.192
yawning	Fatigue Yawn	50	20.00%	0.361	0.120
notdrowsy	Attentive Driving	100	32.00%	0.249	0.254

• **YOLO Distraction Model Specificity: 99.20%** (Only 2/250 false alarms across all face postures).

3. Kaggle State Farm DDD Benchmark (1,065 Images)

Metric	Score	Significance
Total Evaluated Images	1,065	332 phone infractions + 733 cups/steering controls
Phone Sensitivity (Recall)	99.70%	Caught 331 out of 332 out-of-domain phones
Mean Phone Confidence	0.715 (71.5%)	Strong detection confidence
Optimal Operating Threshold	conf = 0.65	Yields 92.0% recall with 68.7% F1-score